=== UNIT 4: MULTIVARIATE DIFFERENTIAL CALCULUS, JACOBIANS & EXTREMA ===

>> 4.1 Partial Derivatives and Euler’s Theorem on Homogeneous Functions

Engineering systems typically involve multivariable response surfaces where functions depend on several independent physical variables z = f(x, y).
* Partial Derivatives: The partial derivative dz/dx is evaluated by differentiating with respect to x while treating y strictly as a constant. Higher-order mixed partials satisfy Clairaut’s / Schwarz’s Theorem: d^2z / (dx dy) = d^2z / (dy dx) if partial derivatives are continuous.
* Homogeneous Functions: A function f(x, y) is homogeneous of degree n if f(t*x, t*y) = t^n * f(x, y) for all scalars t > 0.
* Euler’s Theorem for Homogeneous Functions:
If u = f(x, y) is a homogeneous function of x and y of degree n having continuous partial derivatives, then:
x * (du/dx) + y * (du/dy) = n * u.
* Second Order Extension of Euler’s Theorem:
x^2 * (d^2u / dx^2) + 2*x*y * (d^2u / (dx dy)) + y^2 * (d^2u / dy^2) = n * (n - 1) * u.

>> 4.2 Total Differential and Chain Rule

* Total Differential (dz): For z = f(x, y), dz = (dz/dx) dx + (dz/dy) dy. Quantifies the first-order linear approximation of change delta z due to small increments delta x and delta y.
* Chain Rule for Composite Functions: If z = f(u, v) where u = phi(x, y) and v = psi(x, y):
dz/dx = (dz/du)*(du/dx) + (dz/dv)*(dv/dx)
dz/dy = (dz/du)*(du/dy) + (dz/dv)*(dv/dy).

>> 4.3 Jacobians and Functional Transformations

* Definition of Jacobian: If u = u(x, y) and v = v(x, y) are two differentiable functions, the Jacobian of (u, v) with respect to (x, y), denoted J = d(u, v) / d(x, y), is the determinant:
J = det([du/dx du/dy ; dv/dx dv/dy]) = (du/dx)*(dv/dy) - (du/dy)*(dv/dx).
* Fundamental Properties of Jacobians:
1. Reciprocal Property: J * J’ = 1, where J = d(u, v)/d(x, y) and J’ = d(x, y)/d(u, v).
2. Chain Rule Property: If u, v are functions of r, s, and r, s are functions of x, y, then d(u, v)/d(x, y) = [d(u, v)/d(r, s)] * [d(r, s)/d(x, y)].
3. Functional Dependence Test: Two functions u(x, y) and v(x, y) are functionally dependent (i.e., there exists a relation F(u, v) = 0) if and only if their Jacobian identically vanishes: d(u, v) / d(x, y) = 0.

>> 4.4 Maxima and Minima of Functions of Two Variables

* Critical Points: Points (a, b) where both first partial derivatives vanish simultaneously: fx(a, b) = 0 and fy(a, b) = 0.
* Second Derivative Test: Compute second partial derivatives at (a, b):
r = fxx = d^2f / dx^2, s = fxy = d^2f / (dx dy), t = fyy = d^2f / dy^2.
Discriminant: Delta = r * t - s^2.
1. Local Minimum: If Delta > 0 and r > 0.
2. Local Maximum: If Delta > 0 and r < 0.
3. Saddle Point (Neither Max nor Min): If Delta < 0.
4. Inconclusive Test: If Delta = 0 (higher order investigations required).
* Lagrange's Method of Undetermined Multipliers: To find extrema of f(x, y, z) subject to constraint phi(x, y, z) = 0:
Construct auxiliary Lagrange function L(x, y, z, lambda) = f(x, y, z) + lambda * phi(x, y, z).
Solve the system of 4 equations: dL/dx = 0, dL/dy = 0, dL/dz = 0, dL/d(lambda) = phi(x, y, z) = 0.

[SOLVED PROBLEM 1: Application of Euler's Theorem to Non-Homogeneous Formulation]

Problem: If u = arcsin([x + y] / [sqrt(x) + sqrt(y)]), prove that x*(du/dx) + y*(du/dy) = (1/2) * tan(u).
Solution Step 1: Let z = sin(u) = [x + y] / [sqrt(x) + sqrt(y)].
Solution Step 2: Determine homogeneity and degree of z:
Numerator N(x, y) = x + y is homogeneous of degree 1.
Denominator D(x, y) = sqrt(x) + sqrt(y) is homogeneous of degree 1/2.
z(tx, ty) = [tx + ty] / [sqrt(tx) + sqrt(ty)] = [t*(x + y)] / [t^{1/2}*(sqrt(x) + sqrt(y))]
= t^{1 - 1/2} * z(x, y) = t^{1/2} * z(x, y).
Thus, z is a homogeneous function of x and y of degree n = 1/2.
Solution Step 3: Apply Euler's Theorem to z:
x * (dz/dx) + y * (dz/dy) = n * z = (1/2) * z.
Solution Step 4: Express derivatives in terms of u:
Since z = sin(u), dz/dx = cos(u) * (du/dx) and dz/dy = cos(u) * (du/dy).
Substitute: x * [cos(u) * (du/dx)] + y * [cos(u) * (du/dy)] = (1/2) * sin(u).
Factor cos(u): cos(u) * [x*(du/dx) + y*(du/dy)] = (1/2) * sin(u).
Divide by cos(u): x*(du/dx) + y*(du/dy) = (1/2) * [sin(u) / cos(u)] = (1/2) * tan(u).
(Hence Proved).

[SOLVED PROBLEM 2: Jacobian Coordinate Transformation (Polar Coordinates)]

Problem: For the Cartesian to polar transformation x = r*cos(theta), y = r*sin(theta), calculate J = d(x, y)/d(r, theta) and J' = d(r, theta)/d(x, y), and verify J * J' = 1.

Solution Step 1: Compute partial derivatives for J:

dx/dr = cos(theta), dx/d(theta) = -r*sin(theta)

dy/dr = sin(theta), dy/d(theta) = r*cos(theta).

Solution Step 2: Evaluate determinant J:

J = det([cos(theta) -r*sin(theta) ; sin(theta) r*cos(theta)])

J = cos(theta) * [r*cos(theta)] - [-r*sin(theta)] * sin(theta) = r*cos^2(theta) + r*sin^2(theta) = r.

Solution Step 3: Compute inverse relations:

r = sqrt(x^2 + y^2), theta = arctan(y / x).

dr/dx = x / sqrt(x^2 + y^2) = x / r = cos(theta).

dr/dy = y / sqrt(x^2 + y^2) = y / r = sin(theta).

d(theta)/dx = [1 / (1 + (y/x)^2)] * (-y / x^2) = -y / (x^2 + y^2) = -sin(theta) / r.

d(theta)/dy = [1 / (1 + (y/x)^2)] * (1 / x) = x / (x^2 + y^2) = cos(theta) / r.

Solution Step 4: Evaluate determinant J':

J' = det([cos(theta) sin(theta) ; -sin(theta)/r cos(theta)/r])

J' = [cos(theta) * cos(theta)/r] - [sin(theta) * (-sin(theta)/r)] = (cos^2 + sin^2) / r = 1 / r.

Solution Step 5: Verification of Reciprocal Property:

J * J' = r * (1 / r) = 1. (Verified).

[SOLVED PROBLEM 3: Complete Classification of Multivariable Extrema]

Problem: Find and classify all critical points of the function f(x, y) = x^3 + y^3 - 3*x - 12*y + 20.

Solution Step 1: Compute first partial derivatives and set to zero:

fx = 3*x^2 - 3 = 0 => 3*(x^2 - 1) = 0 => x = +1 or x = -1.

fy = 3*y^2 - 12 = 0 => 3*(y^2 - 4) = 0 => y = +2 or y = -2.

There are 4 distinct critical points: P1(1, 2), P2(1, -2), P3(-1, 2), P4(-1, -2).

Solution Step 2: Compute second partial derivatives:

r = fxx = 6*x, s = fxy = 0, t = fyy = 6*y.

Discriminant: Delta = r*t - s^2 = (6*x)*(6*y) - 0 = 36*x*y.

Solution Step 3: Classify each point:

1. Point P1(1, 2): Delta = 36*(1)*(2) = +72 > 0. r = 6*(1) = +6 > 0. Conclusion: LOCAL MINIMUM.

Minimum Value: f(1, 2) = 1 + 8 - 3 - 24 + 20 = 2.

2. Point P2(1, -2): Delta = 36*(1)*(-2) = -72 < 0. Conclusion: SADDLE POINT.

3. Point P3(-1, 2): Delta = 36*(-1)*(2) = -72 < 0. Conclusion: SADDLE POINT.

4. Point P4(-1, -2): Delta = 36*(-1)*(-2) = +72 > 0. r = 6*(-1) = -6 < 0. Conclusion: LOCAL MAXIMUM.

Maximum Value: f(-1, -2) = -1 - 8 + 3 + 24 + 20 = 38.

[SOLVED PROBLEM 4: Constrained Optimization via Lagrange Multipliers]

Problem: A rectangular open-top water reservoir of volume V = 32 m^3 is to be constructed. Determine the optimal dimensions (length x, width y, depth z) that minimize the total surface area of material required.

Solution Step 1: Formulate objective function and constraint:

Surface Area to minimize (open top): S(x, y, z) = x*y + 2*x*z + 2*y*z (base + 4 sides).

Constraint: Volume phi(x, y, z) = x*y*z - 32 = 0.

Solution Step 2: Set up Lagrange function: L(x, y, z, lambda) = x*y + 2*x*z + 2*y*z + lambda * (x*y*z - 32).

Solution Step 3: Compute partial derivatives:

dL/dx = y + 2*z + lambda*y*z = 0 => lambda = - (y + 2*z) / (y*z) = - 1/z - 2/y.

dL/dy = x + 2*z + lambda*x*z = 0 => lambda = - (x + 2*z) / (x*z) = - 1/z - 2/x.

dL/dz = 2*x + 2*y + lambda*x*y = 0 => lambda = - (2*x + 2*y) / (x*y) = - 2/y - 2/x.

Solution Step 4: Equate lambda expressions:

- 1/z - 2/y = - 1/z - 2/x => 2/y = 2/x => x = y.

Equate first and third: - 1/z - 2/x = - 2/x - 2/x => 1/z = 2/x => x = 2*z (or z = x/2).

Solution Step 5: Substitute into volume constraint:

x * y * z = x * x * (x / 2) = 32 => x^3 / 2 = 32 => x^3 = 64 => x = 4 m.

Then y = x = 4 m, and z = x / 2 = 2 m.

Optimal Dimensions: Length = 4 meters, Width = 4 meters, Depth = 2 meters.

Minimum Material Area: S = 4*4 + 2(4)(2) + 2(4)(2) = 16 + 16 + 16 = 48 m^2.

>> 4.5 University Review & Examination Questions

* Q1: State and prove Euler’s Theorem for a homogeneous function u = f(x, y) of degree n.

Answer: By definition of homogeneity, f(t*x, t*y) = t^n * f(x, y). Differentiating both sides with respect to parameter t using the multivariable chain rule: [df/d(t*x)] * [d(t*x)/dt] + [df/d(t*y)] * [d(t*y)/dt] = n * t^{n-1} * f(x, y). Since d(t*x)/dt = x and d(t*y)/dt = y, this gives x * fx(t*x, t*y) + y * fy(t*x, t*y) = n * t^{n-1} * f(x, y). Setting t = 1 yields the standard theorem: x * fx(x, y) + y * fy(x, y) = n * f(x, y).

* Q2: Explain the necessary and sufficient conditions for a function of two variables f(x, y) to have an extreme value at (a, b).

Answer: Necessary condition: The first partial derivatives must vanish: fx(a, b) = 0 and fy(a, b) = 0. Sufficient condition: Let r = fxx(a, b), s = fxy(a, b), t = fyy(a, b), and Delta = r*t - s^2. (1) If Delta > 0 and r > 0, f has a strict Local Minimum at (a, b). (2) If Delta > 0 and r < 0, f has a strict Local Maximum at (a, b). (3) If Delta < 0, (a, b) is a Saddle Point. (4) If Delta = 0, the second-derivative test fails and higher-order terms must be investigated.

>> 4.6 Key Takeaways and Revision Points

* Euler’s theorem: x*(du/dx) + y*(du/dy) = n*u for homogeneous functions of degree n.
* Jacobians J = d(u,v)/d(x,y) measure local area stretching under coordinate transformations.
* Extremum classification discriminant: Delta = r*t - s^2. Delta > 0 gives extrema (sign of r determines min/max); Delta < 0 gives saddle.
* Lagrange multipliers solve constrained optimization problems by identifying parallel gradient vectors (grad f = -lambda * grad phi).